

Satya Sai Vasamsetty

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TECHNICAL SKILLS

Languages	Python, C/C++, SQL, JavaScript, Bash Scripting
ML / CV	PyTorch, OpenCV, YOLO (v5/v8/World), Person Re-ID, Facial Recognition, ONNX, NCNN, Albumentations
Gen AI / LLMs	Qwen VLM, Gemini Multimodal, RAG (Retrieval-Augmented Generation), LangChain, LLM Fine-tuning, Prompt Engineering, Agentic AI, Multimodal AI, Rate Limiting, Load Balancing
Infra & Tools	Docker, AWS, Linux, Git, Nginx, Raspberry Pi, Node.js, React
Databases	MySQL, MongoDB, Redis
Concepts	Edge Deployment, Object Detection & Tracking, Generative AI Pipelines, Vector Embeddings, REST APIs, Networking, OS Fundamentals

EXPERIENCE

Senior Deep Learning Engineer January 2024 – Present
Tarsyer Insights LLP Pune, Maharashtra, India

- Delivered 9 production Computer Vision and Generative AI systems across 1,308+ locations for clients, Trent, Royaloak, Dr Batras, Mahindra Logistics, Ultratech Cement, Oerlikon Balzers, Maruti Suzuki, and ArcelorMittal.
- Managed 1,000+ edge devices via a central server with zero-data-loss MySQL pipelines, designed a Git + Python architecture for fleet-wide software versioning and deployment.
- Built and self-hosted open-source Generative AI inference (Qwen LLM) with custom rate limiters and load balancers, designed agentic workflows for multi-step reasoning and structured output generation.
- Led full project lifecycle, client requirement gathering, model selection, data pipeline design, edge deployment, and production monitoring.

PROJECTS

Retail Store Production Monitoring Trent Limited (Westside & Zudio)
YOLO, Person Re-ID, Qwen VLM, Gemini, Generative AI, Facial Recognition, ResNet

- Deployed across **1,200 stores** on 1,000+ edge devices, 10 use-cases, footfall (M/F/child), dwell time, fire-exit blockage, crowd alerts, store manager tracking, intrusion detection, and back-office cleanliness via Generative AI (multimodal LLM).

Customer Analytics System Royaloak Furniture
MobileNet, Object Tracking, Person Re-ID, ResNet, Cosine Distance

- Accurate group and unique customer counting across **86 stores**, re-identification eliminates duplicate counts across frames for reliable footfall KPIs.

Patient & Operations Monitoring Dr Batras
YOLO, NCNN, Person Re-ID, Gemini, Generative AI, Audio Analysis

- 10 use-cases across **11 clinics**, patient flow, dwell time, doctor-patient time, staff uniform compliance, greeting check, audio interaction analysis via Generative AI (Gemini multimodal LLM), and real-time device health alerts.

Warehouse & Dock Monitoring Mahindra Logistics
YOLO, MobileNet, Gemini, Generative AI, YOLO-World, CLIP, Clustering

- 7 use-cases across **3 hubs** (Pune, Jaipur, Hyderabad), dock activity, truck number-plate detection, loitering alerts, pallet counting, mobile-phone usage detection via Generative AI (Gemini multimodal), and suspicious-activity monitoring.

Star Bazaar Store Monitoring Trent Limited
YOLO, Binary Classifier, ONNX, ResNet Re-ID, Facial Recognition

- 6 use-cases across **3 sites**, footfall, shelf empty-box detection for fruits/vegetables/staples (binary classifier, ONNX), fire-exit monitoring, and cash till activity.

Tool Counting & Measurement Oerlikon Balzers
YOLO, Albumentations, PCA, Linear Regression, Raspberry Pi

- Counts 10–600 tools on a metallic tray at **99.98% accuracy**, measures diameter (± 0.5 mm) and length (± 1 mm) in <1 s on Raspberry Pi. Reduced manual work by **90%**. Trained on 249 images with custom augmentation.

Warehouse Production Monitoring Ultratech Cement
MobileNet, Object Tracking, Object Detection

- Automated cement bag counting (entry & exit) across **2 production sites**, eliminating manual tracking and improving inventory accuracy.

Activity Assessment System Maruti Suzuki
Object Detection, Color Segmentation, Motion Detection, Contour Detection, Sequence Logic

- CV system monitoring 10 assembly activities at **99.7% accuracy**, tracks task duration, idle time, and step transitions for trainer performance evaluation.

Crane Hook Safety Detection ArcelorMittal
YOLOX Nano, ONNX Runtime, ZeroMQ, Raspberry Pi, Moxa PLC, RTSP

- Safety-critical real-time CV system classifying crane ladle hook states (hooked / mishooked / active) and triggering PLC signals (safe / unsafe / warning) to prevent accidents during molten metal transfers, multi-camera consensus with gated 10-second hold before any signal fires.

EDUCATION

Jawaharlal Nehru Technological University Aug 2019 – July 2023
Bachelor of Technology in Computer Science Sangareddy, Telangana, India

ACTIVITIES & ACHIEVEMENTS

- Volunteered as Python and Web Development mentor at College Technical Fest.
- Conducted an interactive Git & GitHub workshop for students at College Technical Fest.
- CODE ADDICTS, created and hosted a programming contest at YUGA2K22 on HackerRank.